

Zeyu Ding

Research Interests: Differential Privacy, Database,
Privacy-Preserving Machine Learning, Formal Methods



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Recent Positions

- 2022 – **Binghamton University** Binghamton, NY
now Assistant Professor, School of Computing
- 2021 **Meta** Menlo Park, CA
Research Scientist, Facebook Reality Labs

Education

- 2022 **Penn State University** University Park, PA
PhD, Computer Science
- 2006 **Zhejiang University, Chu Kochen Honors College** Hangzhou, China
BS, Mathematics

Honors and Awards

- 2021 Best Paper Award Runner-Up, ACM Conference on Computer and Communications Security
- 2021 Graduate Outstanding Teaching Award, Penn State University
- 2020 Best Paper Award Runner-Up, ACM Conference on Computer and Communications Security
- 2019 The Caspar Bowden Award for Outstanding Research in PET Runner-Up
- 2019 Graduate Outstanding Research Award, Penn State University
- 2018 Best Paper Award, ACM Conference on Computer and Communications Security

Grants

PI, “Collaborative Research:SaTC:CORE:Medium: Differentially Private SQL with Flexible Privacy Modeling, Machine-Checked System Design and Accuracy Optimization,” National Science Foundation, Award #2317233, \$335,584, 2024–2028.

Publications

- 2026 Guanlin He, Yingtai Xiao, Jiamu Bai, Xin Gu, Zeyu Ding, Wenpeng Yin, and Daniel Kifer. Accurate and Scalable Matrix Mechanisms via Divide and Conquer. *arXiv:2604.00868*.
- 2026 Andrew Cascio, KinChin Tong, Daniel Kifer, Zeyu Ding, and Danfeng Zhang. DP4SQL: Differentially Private SQL with Flexible Privacy Policies. *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*.
- 2026 Guanlin He, Yingtai Xiao, Levent Toksoz, Zeyu Ding, Danfeng Zhang, and Daniel Kifer. ResidualPlanner+: A Scalable Matrix Mechanism for Marginals and Beyond. *The International Journal on Very Large Data Bases*.
- 2026 Vrushket More, Lyra Lu, Zeyu Ding, Zhaohan Xi, Seth Mizia, and Nancy L. Guo. TheraMind: A Multi-LLM Ensemble for Accelerating Drug Repurposing in Lung Cancer via Case Report Mining. *npj Precision Oncology*.

- 2025 Zeyu Ding, John Durrell, Daniel Kifer, Prottay Protivash, Guanhong Wang, Yuxin Wang, Yingtai Xiao, and Danfeng Zhang. Avoiding Floating-Point Side Channels in the Report Noisy Max with Gap Mechanism. *Journal of Privacy and Confidentiality*.
- 2025 Prem Bhajaj, Saiprakash Nalubolu, Bhargavram Gurram, Mansi Srinivas, Rakesh Choudhary, Weiying Dai, Ling Wang, Tamara Jamaspishvili, Zeyu Ding, and Nancy L Guo. ClinSegAI: A Post-Processing Framework for Superior Histopathology Segmentation Accuracy, Radiomics Feature Preservation, and Quantitative Analysis. *Computers in Biology and Medicine*.
- 2024 Prottay Protivash, John Durrell, Zeyu Ding, Danfeng Zhang, and Daniel Kifer. Reconstruction Attacks on Aggressive Relaxations of Differential Privacy. *Journal of Privacy and Confidentiality*.
- 2023 Zeyu Ding, Yuxin Wang, Yingtai Xiao, Guanhong Wang, Danfeng Zhang and Daniel Kifer. Free Gap Estimates from the Exponential Mechanism, Sparse Vector, Noisy Max and Related Algorithms. *The International Journal on Very Large Data Bases*.
- 2021 Yuxin Wang, Zeyu Ding, Yingtai Xiao, Daniel Kifer and Danfeng Zhang. DPGen: Automated Program Synthesis for Differential Privacy. *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*. **Best Paper Award Runner-Up**
- 2021 Zeyu Ding, Daniel Kifer, Sayed M. Saghaian N. E., Thomas Steinke, Yuxin Wang, Yingtai Xiao and Danfeng Zhang. The Permute-and-Flip Mechanism is Identical to Report-Noisy-Max with Exponential Noise. *arXiv:2105.07260*.
- 2021 Yingtai Xiao, Zeyu Ding, Yuxin Wang, Danfeng Zhang and Daniel Kifer. Optimizing Fitness-For-Use of Differentially Private Linear Queries. *Proceedings of the VLDB Endowment*.
- 2021 Zeyu Ding and Xiao Xiao. Dimensions of Automorphism Group Schemes of Finite Level Truncations of F-cyclic F-crystals. *Transactions of American Mathematical Society*.
- 2020 Yuxin Wang, Zeyu Ding, Daniel Kifer and Danfeng Zhang. CheckDP: An Automated and Integrated Approach for Proving Differential Privacy or Finding Precise Counterexamples. *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*. **Best Paper Award Runner-Up**
- 2020 Zeyu Ding, Yuxin Wang, Danfeng Zhang and Daniel Kifer. Free Gap Information from the Differentially Private Sparse Vector and Noisy Max Mechanisms. *Proceedings of the VLDB Endowment*.
- 2019 Yuxin Wang, Zeyu Ding, Guanhong Wang, Daniel Kifer and Danfeng Zhang. Proving Differential Privacy via Shadow Execution. *The ACM SIGPLAN Conference on Programming Language Design and Implementation*.
- 2018 Zeyu Ding, Yuxin Wang, Guanhong Wang, Danfeng Zhang and Daniel Kifer. Detecting Violations of Differential Privacy. *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*. **Best Paper Award**

Teaching

Binghamton University

CS471: Programming Languages, Spring 2023/2024/2025/2026
 CS561: Topics in Data Privacy, Fall 2022/2023/2024/2025
 CS571: Programming Languages, Spring 2026
 MATH221: Calculus I, Spring 2011/2013, Fall 2008/2012
 MATH222: Calculus II, Spring 2012/2013/2014
 MATH323: Calculus III, Fall 2011
 MATH304: Linear Algebra, Spring 2009/2010/2015, Fall 2009/2015

MATH314: Discrete Math, Fall 2014, Spring 2015
MATH330: Number Theory, Fall 2014
MATH371: Ordinary Differential Equations, Spring 2014

Penn State University

CMPSC461: Programming Language Concepts, Head TA, Fall 2020, Spring 2021

Professional Service

Program Committee

2027 The 27th Privacy Enhancing Technologies Symposium (PETS)
43rd IEEE International Conference on Data Engineering (ICDE)
2026 IEEE European Symposium on Security and Privacy (EuroS&P)
Theory and Practice of Differential Privacy (TPDP) Workshop
2024 Theory and Practice of Differential Privacy (TPDP) Workshop
2023 ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL) SRC
2022 ACM Conference on Computer and Communications Security (CCS)

Grant Review Panel

DOE Advancements in Artificial Intelligence for Science, 2024

External Reviewer

CCS, VLDB, NeurIPS, ICLR, ICML, LICS, TDSC, TPDS, ALGO, JPC, TPDP

School of Computing

Tenure-Track Faculty Search Committee, 2022–23, 2023–24
Graduate Application Review Committee, 2023–24, 2024–25
Graduate Committee, 2025–26